A packing model of a closed universe: random sequential adsorption of unit-rapidity worldlines on a comoving torus

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Abstract

We define and study a model universe built from nothing but mutually avoiding worldlines. Each worldline is a closed path over a conformal time loop $z \in (0, \pi)$, expressed per spatial axis as a sinusoidal expansion whose non-trivial frequency components jointly obey a unit *rapidity* constraint $\sum_{k\geq 2} |k\, a_k| = 1$ (a built-in speed of light), while the frequency-1 amplitude is a free comoving coordinate — a separation that makes homogeneity a property of the measure rather than an assumption. Comoving space is a torus: a path leaving through one face re-enters through the opposite one, so there is no boundary and no statable center. Even frequencies carry a free phase; odd frequencies are shown to already possess their full phase freedom. Packing these worldlines by random sequential adsorption under an everywhere-in-time exclusion rule produces a jam whose count grows as a clean power law $N_{\text{sat}} \sim T^D$ in the time resolution T, with $D = 2.33$ in three spatial dimensions and 1.38–1.40 in two — strictly below the spatial dimension, yet demonstrably not the fractal dimension of any static set: the equal-time slices of the jam are *exactly* uniform (wrapped correlation dimension 3.01 and 2.03; rms comoving spread flat at $\sqrt{1/3}$ over the entire loop), and the worldline bundle's local dimension sweeps from 1 to d with no plateau at D. The model makes a parameter-free kinematic prediction: mover velocities at maximum expansion should follow the arcsine law $P(v) = 2/\pi\sqrt{1 - v^2}$, diverging integrably at the speed cap. The packed state inherits the law's shape — including the relativistic pile-up at $v \to 1$ — but with a measurable tilt toward slow movers ($\langle v^2 \rangle = 0.40$ vs the proposal's 0.50): jamming *selects on phase*. The resulting equation of state at the turnaround is matter-like, $w = 0.070$, flat across the entire resolution ladder and identical under two independent energy dictionaries. We close with the geometry of what remains unexplored — multi-frequency worldlines, intersecting "unique groups" and their subpaths — and with a discussion of whether the jam supports a strange-attractor reading.

1 Introduction

Most model universes start from fields on a background. This one starts from *paths*. The only objects in the model are worldlines — closed trajectories through a finite expanding-and-contracting spacetime — and the only law is that they may not intersect. "Matter" is whatever set of worldlines survives being packed, one at a time and at random, until nothing more fits. The questions one can ask of such a universe are correspondingly concrete. How much fits? What does the surviving population look like — where is it, how fast does it move, what microscopic traits did the packing select for? And do any of the answers resemble physics?

The packing problem itself belongs to a classical family: random sequential adsorption (RSA), the irreversible random placement of mutually excluding objects, exactly solvable in one dimension [1] and much studied since for compact shapes [2, 3]. Our objects are not compact. A worldline must avoid every other accepted worldline at *every* moment of the loop simultaneously, so the excluded volume seen by a candidate is a joint, global object — and this, it will turn out, is where all of the model's nontrivial structure lives.

This paper does three things. First, it defines the model carefully (Section 2): the worldline family, the rapidity constraint that plays the role of a speed of light, the toroidal comoving space that removes any preferred center, the phase freedom of even frequencies, and the intersection rules — including the notions of *unique groups* and *subpaths*, which define how the model is meant to be filled beyond the sector explored here. Second, it reports what has been measured in the first explored sector — worldlines with a single wiggle frequency per axis, one path per group — packed at scale on GPU hardware (Sections 3–7): the growth exponent of the jam, the exact homogeneity of its slices, the absence of a geometric carrier for the exponent, the parity structure of the surviving frequencies, an arcsine velocity law with a measured deviation that quantifies how jamming selects on phase, and a resolution-independent, dictionary-robust equation of state. Third, it discusses what the results do and do not license one to conclude — including the question of whether the jam is usefully described as a strange attractor — and lays out the unexplored territory (Section 8).

We have tried to keep the presentation self-contained and readable rather than maximally terse; where a derivation is short we give it, and where a claim is speculative we say so. All code (packing engines, orchestration, analysis), all data, and a dated lab notebook are public.[1]

2 The model

2.1 Worldlines as constrained sinusoidal expansions

Time is a coordinate $z \in (0, \pi)$, and the universe carries a scale factor $a(z) = \sin z$: it expands from a bang at $z = 0$ to a maximum at the turnaround $z = \pi/2$ and recontracts to a crunch at $z = \pi$. Along each spatial axis, a worldline is a superposition of integer-frequency sine terms,

$$x(z) \;=\; a_1 \sin z \;+\; \sum_{k\geq 2} a_k \left[\sin(kz + f_k) - \sin f_k\right], \qquad k \in \mathbb{Z}, \tag{1}$$

subject to three structural rules.

Closure. The path must begin and end at zero: every worldline meets the bang and the crunch. The subtracted constants $\sin f_k$ in Eq. (1) enforce this in the presence of phases — each component is re-pinned to zero at the endpoints. Closure is also why phases are restricted as they are below.

The rapidity constraint. All frequency components except $k = 1$ jointly obey

$$\sum_{k\geq 2} |k\, a_k| \;=\; 1. \tag{2}$$

[1] https://github.com/universe-analysis/universe-generator.

Since each term contributes at most $|ka_k|$ to $|dx/dz|$, this is a unit speed budget — a speed of light built into the family rather than imposed on it. A worldline with $0.5 \sin 2z$ spends its whole budget ($0.5 \times 2 = 1$); one with $\frac{1}{5} \sin 3z + \frac{1}{5} \sin 2z$ spends $\frac{3}{5} + \frac{2}{5} = 1$ likewise. The amplitudes of all non-trivial components are thus dependent on one another: speed spent on one frequency is unavailable to the rest.

The free comoving coordinate. The $k = 1$ amplitude is exempt from the budget and drawn freely, $a_1 \in (-1, 1)$. This term has a special meaning: in *comoving* coordinates $X = x/\sin z$ it is simply the constant a_1 — a static comoving position, the Hubble flow. Separating it from the rapidity constraint is what erases the preferred center: position costs nothing, so a uniform draw of a_1 is exactly the homogeneous measure, and nothing in the dynamics couples where a worldline sits to how fast it may move. (Section 5 shows the packing preserves this homogeneity exactly.)

Phases. Even frequencies may be freely modulated by a constant phase $f_k \in [0, \pi)$. Odd frequencies carry no explicit phase — and need none: for odd k, a sign flip of the amplitude is exactly a phase shift by π ($\sin(kz + \pi) = -\sin kz$ and the closure offset transforms consistently), and this is the entire phase freedom compatible with closure, since $\sin(k\pi + f) = \pm \sin f$ cancels the re-pinning constant only when k is even. Odd components are, in this sense, already modulated to the highest allowable degree; even components acquire a genuinely new continuous degree of freedom. This asymmetry between the parities threads through everything that follows.

The frequencies k range over all positive integers in the abstract model. Under discrete analysis (Section 2.5) the timestep resolution imposes a Nyquist ceiling, which is the only place a frequency cutoff enters.

2.2 Comoving space is a torus

The bounds of the universe along each axis are the curves $\pm \sin z$ — the envelope traced by the extremal $k = 1$ worldlines. A path that crosses $+\sin z$ does not end or reflect: it re-enters at $-\sin z$ moving toward the center it just left. In comoving coordinates this is the statement that X lives on a circle of circumference 2, with $X = +1$ and $X = -1$ the same point; with d spatial axes, comoving space is a d-torus. Distances (and hence intersections) are computed in the minimum image. Two structures that a bounded model would possess are thereby not removed but rendered *unstatable*: there is no wall, and no point is distinguishable as a center.

2.3 The spacetime view

A full worldline in d+1 dimensions is the parametric curve

$$\big(x_1(z),\ \ldots,\ x_d(z),\ z\big), \qquad z \in (0, \pi),$$

with an independent expansion (1) per axis: a d-dimensional path unfolding over time. All results in this paper are for $d = 2$ and $d = 3$ (“2+1” and “3+1”). It is worth noting what the family (1) is, seen whole: since every closed path with bounded velocity admits a sine expansion, the model with all frequencies admitted is (a discretization of) the space of *all* closed unit-rapidity paths on the loop — the single-frequency sector studied below is its first, sparsest slice.

2.4 Intersections, unique groups, and subpaths

Two worldlines *intersect* if they come within the exclusion distance of each other (made precise in Section 2.5) at any moment of the loop. The organizing notion of the model is the *unique group*: a set of paths that each intersect at least one other member of their own group and never intersect any path of another group. A unique group with several member paths is one object that exists in several places at once; a group with a single path is the degenerate case.

The intended construction of a full universe proceeds in two stages. First, fill the loop with mutually non-intersecting paths — every group a singleton — until jamming (or any chosen earlier stopping point). Second, continue filling by adding *subpaths*: new paths that intersect a member of an existing group (thereby joining it) while still avoiding all other groups. The first stage is a hard-sphere-like packing problem; the second redistributes the remaining room among the existing identities rather than creating new ones.

Everything measured in this paper concerns the first stage. Subpaths, and worldlines carrying more than one budget frequency, are defined by the model but not yet explored; we return to what questions they open in Section 8, and we deliberately draw no conclusions about them here.

2.5 Discrete analysis: timesteps, Nyquist, exclusion

Analysis proceeds at a chosen *resolution* T: the loop is sampled at T evenly spaced interior times $z_i = i\pi/(T{+}1)$, $i = 1, \ldots, T$. The resolution fixes everything else:

- **Nyquist ceiling.** Frequencies are admitted up to $k \le T$ — beyond that a component cannot be resolved by the grid.
- **Exclusion.** Each path carries an exclusion radius of $1/T$ in comoving coordinates (Chebyshev metric, minimum-image on the torus): two paths intersect if at any sampled time their comoving separation is below $2/T$ per axis. The exclusion cell shrinks exactly as the resolution grows, so "how much fits at resolution T" is a well-posed scaling question.

Packing is random sequential adsorption: candidates are drawn from the measure described above (uniform a_1; frequencies from the admitted band with a high-frequency bias and pairwise-coprime axis frequencies; signs and phases uniform) and accepted iff they intersect no previously accepted path. True jamming is logarithmically slow to reach at large T, so runs stop at an acceptance-rate cutoff of 10^{-6} (one accepted candidate per million attempts); the flatness of the measured local scaling slopes across the ladder (Fig. 1, lower panel) indicates the configurations are well inside the scaling regime, and the extrapolation to literal jamming is flagged as an open problem rather than assumed away.

3 Generation at scale

The explored sector — one budget frequency per axis, i.e. $x(z) = a\,[\sin(bz + f) - \sin f] + a_1 \sin z$ with $|ab| = 1$ per axis — is packed by CUDA engines (one for each d). The engine's collision test exploits a small identity: the exclusion threshold in physical coordinates is $\sin(z_i) \cdot 2/T$, which in comoving coordinates collapses to the constant $2/T$ at every timestep, so a uniform spatial hash with exactly T cells per axis per timestep gives an $O(NT)$ (rather than $O(N^2T)$) test, wrapping mod T for the minimum image. Memory scales as T^{d+1}; consumer GPUs reach

$T = 200$ at $d = 3$ and $T = 400$ at $d = 2$. Campaigns are dispatched over a small GPU fleet by a resumable orchestrator, and each accepted worldline's parameters (a, b, a_1, f) per axis are dumped, so any configuration can be reconstructed analytically at any z offline. The datasets below are: $d = 3$, $T = 20$–200 in steps of 10, five seeds per T; $d = 2$, $T = 40$–400 in steps of 20, eight seeds per T.

Four estimator families are used. The *jam-count exponent* is the weighted log–log slope of $N_{\rm sat}$ vs T with seed-bootstrap errors, checked for power-law quality via local pairwise slopes. The *correlation dimension* D_2 of an equal-time slice uses the Grassberger–Procaccia integral [4] with both spherical (L^2) and cubic (L^∞) probes; because the domain is periodic, the correct estimator wraps distances (minimum image; implemented with a periodic k-d tree), and we also report the naive open-boundary estimator on the same clouds as a deliberate control. *Box-counting* of slices and of the full worldline bundle is reported as local slope vs scale and used diagnostically (it converges far too slowly in T to serve as a headline estimator). Kinematic and equation-of-state quantities are computed analytically from the dumped parameters (Section 7).

4 How much fits: the jam-count exponent

The jammed count is a clean power law, $N_{\rm sat} \sim T^D$, over the full measured ladder in both dimensions (Fig. 1):

$$D_{3+1} = 2.273{\pm}0.003 \text{ (full ladder)}, \quad 2.331{\pm}0.003\ (T \geq 100); \qquad D_{2+1} = 1.396{\pm}0.002, \quad 1.375{\pm}0.004.$$

Local slopes are flat (scatter 0.066 in 3+1, 0.080 in 2+1) with the mild low-T bend absorbed by the high-T fits; we quote $D \approx 2.33$ and $D \approx 1.38$ as the converged values at this cutoff. At the top of the ladders the jams hold $\langle N_{\rm sat}\rangle \approx 1.6 \times 10^5$ worldlines ($T = 200$, $d = 3$) and 6.1×10^3 ($T = 400$, $d = 2$).

The exponent sits well below the spatial dimension in both cases. Pure spatial packing would give $N_{\rm sat} \sim T^d$: each path excludes $O(1)$ cells of a T^d comoving grid, and slices, as the next section shows, are packed homogeneously. The deficit — $T^{2.33}$ against a T^3 ceiling — is therefore not a matter of where the worldlines sit. It is the price of the *joint* constraint: a candidate must thread the gaps of every one of T time slices simultaneously, and the number of trajectories able to do so grows more slowly than the room available in any single slice. We call D a *packing-number exponent*, and Section 6 substantiates the name by showing no static set carries it as a geometric dimension.

5 Where it sits: exact homogeneity

The matter distribution of the jam is exactly uniform, and this holds in two senses. Across space: the wrapped correlation dimension of the turnaround slice converges onto the space-filling value with the probe-shape gap collapsed,

$$D_2 = 3.019 \text{ (sphere) } / \ 3.010 \text{ (cube)} \quad (d = 3), \qquad D_2 = 2.032\ /\ 2.025 \quad (d = 2),$$

seed-averaged over $T \geq 100$ (Fig. 2). Across time: fixing one universe and scanning every conformal time, the rms comoving spread sits at $\sqrt{1/3} \approx 0.5774$ — the exact value for a uniform torus — flat across the entire loop, with the slice box-counting dimension confined to a

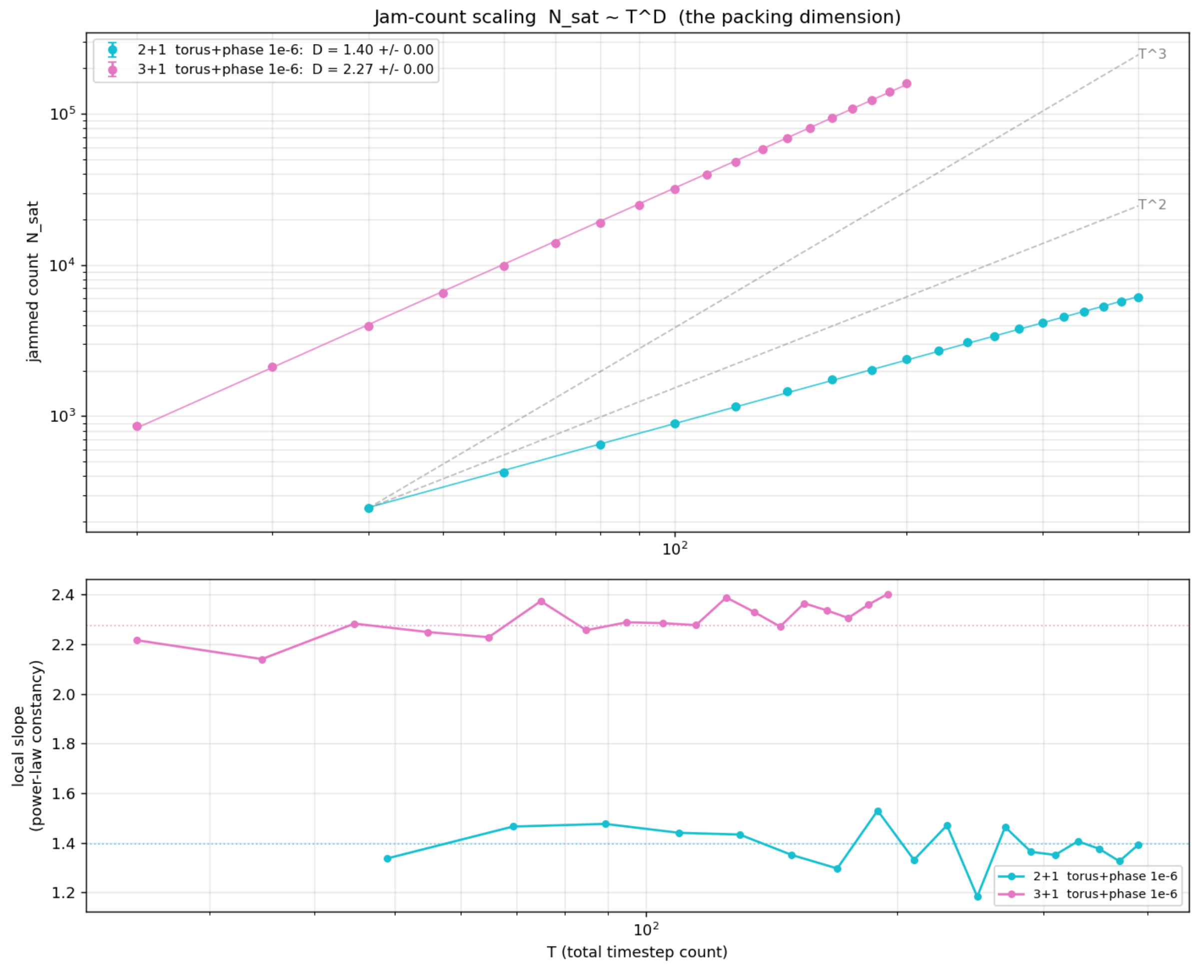

Figure 1: Jam count N_{sat} vs resolution T (log–log) for the two explored dimensions, with T^2 and T^3 guides. Lower panel: local pairwise slopes — flat across the ladder, i.e. clean power laws with no crossover. Fitted exponents: $D = 2.27$ (full ladder) rising to 2.33 on the $T \geq 100$ half in 3+1; 1.40 full / 1.38 high-T in 2+1.

narrow band (Fig. 3; its residual deficit from 3 is the generic slow convergence of box-counting, not structure).

Homogeneity of the *proposal* measure is built in — that is what the free comoving coordinate is for. The content of the measurement is that *jamming preserves it*: the exclusion interaction thins the population by orders of magnitude without imprinting any density structure on the survivors at any measurable scale, at any time. The bang/crunch collapse lives entirely in the physical coordinates ($x = X \sin z \to 0$); the comoving arrangement never knows it is happening.

One methodological remark, since it is easy to get wrong: the naive correlation estimator (unwrapped positions, open-boundary distances) applied to these same configurations reads $D_2 \approx 2.89$ ($d = 3$) and ≈ 1.93 ($d = 2$) with a spurious probe-shape split — a boundary artifact, reproduced deliberately in Fig. 2 as the dotted control. On periodic data the wrapped estimator is not a refinement but the definition; a uniform control cloud returns the exact ambient dimension under it, for both probes.

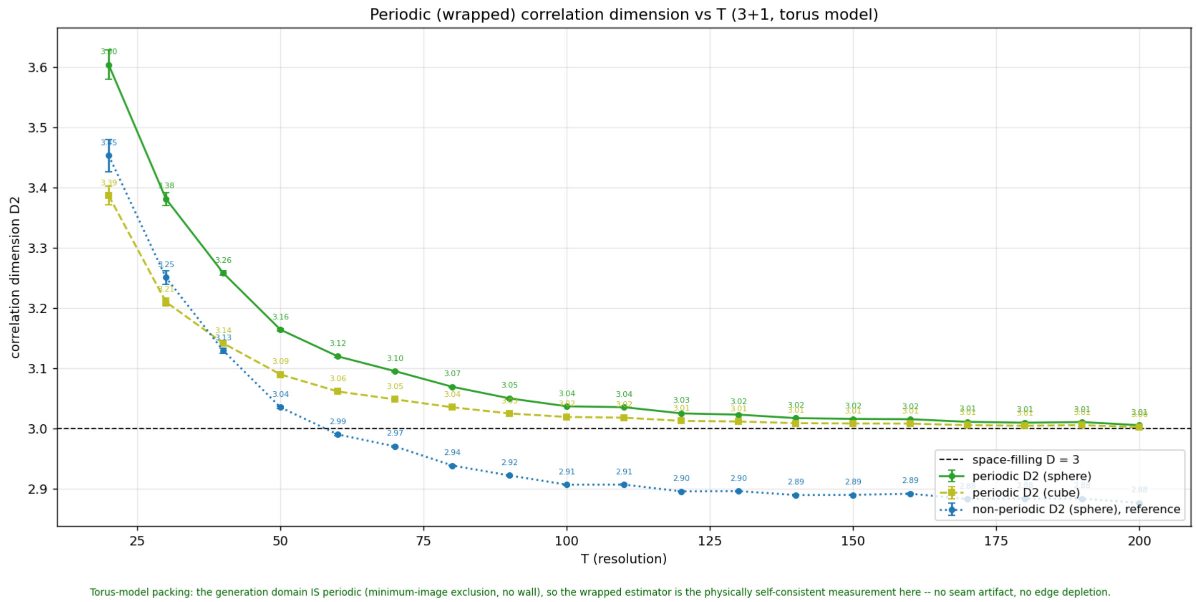

Figure 2: Wrapped (minimum-image) correlation dimension of the turnaround slice vs T (3+1, five seeds per T): spherical and cubic probes converge together onto the space-filling $D = 3$ line (3.01 at $T = 200$). The open-boundary estimator on the same clouds (dotted) reads ∼2.89 — the artifact a non-periodic estimator commits on periodic data, shown as a control.

6 What the exponent is not: no geometric carrier

Given a sub-dimensional exponent, the natural reflex is to look for a fractal — some static set whose box or correlation dimension equals 2.33. The slices cannot be it: they are exactly three-dimensional (Section 5). The remaining candidate is the worldline bundle as a whole, and it fails too (Fig. 4): pooling all timesteps of a jam into one point cloud, the local box-counting dimension runs from ≈1.3 at the finest resolved scale — where the cloud looks like a union of smooth one-dimensional curves — up to ≈2.9 at coarse scales, where it is space-filling. It crosses 2.33 in passing, with no plateau; the same holds treating time as a fourth axis.

For packings of *points*, packing number and box dimension coincide, which is exactly why the reflex exists. For packings of extended curves under a joint across-time constraint the two notions separate, and this model separates them cleanly: geometry says "uniform curves filling the space," counting says "$T^{2.33}$ of them fit." The exponent measures a *dynamical* capacity — how the number of globally non-intersecting threadings scales with resolution — and appears to have no static shadow. We are not aware of a standard name for this quantity in the RSA literature.

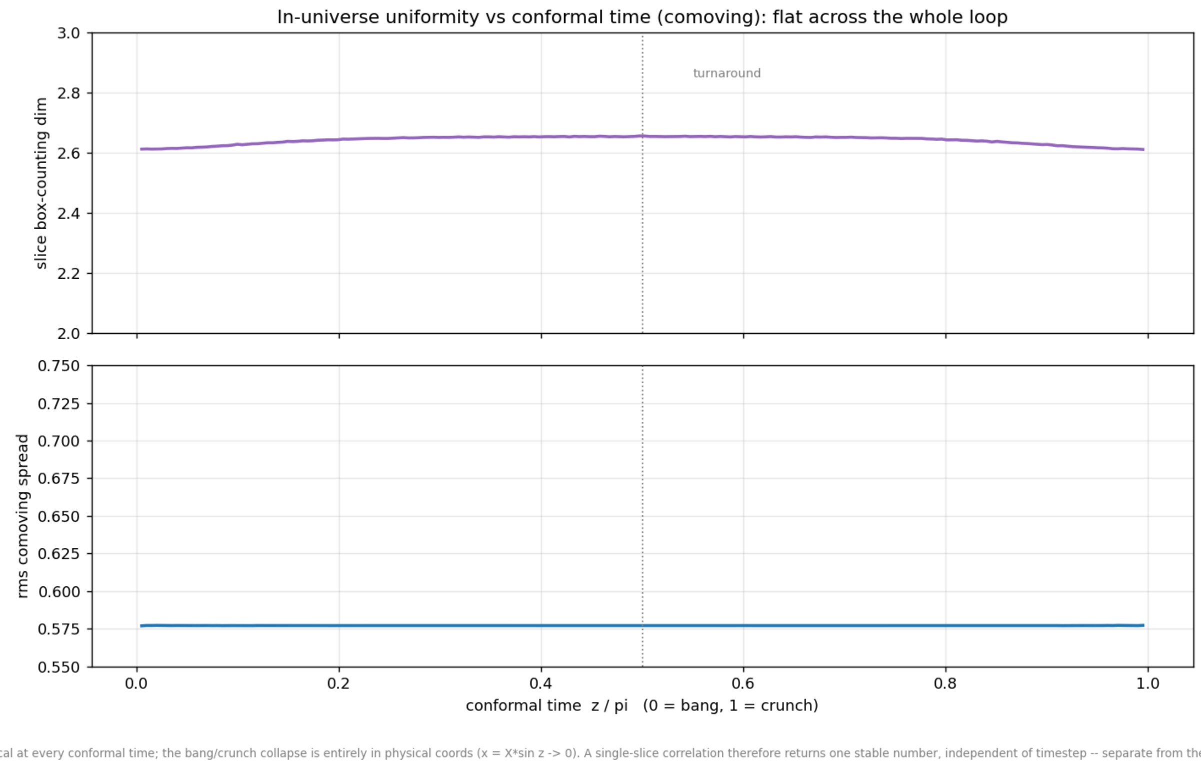

Figure 3: One universe over its whole history (3+1, $T = 200$, five seeds): slice box-counting dimension (top) and rms comoving spread (bottom) vs conformal time. The spread sits on $\sqrt{1/3}$ — the exact uniform-torus value — flat through bang, turnaround, and crunch.

7 What survives: parity, the arcsine law, and the equation of state

7.1 Parity structure of the surviving frequencies

The comoving wiggle $\sin(bz + f)/\sin z$ is symmetric about the turnaround when b is odd (the worldline sits at a velocity extremum there — at rest) and antisymmetric when b is even (it swings through at speed). The jam has a definite, resolution-stable composition along this axis (Fig. 5): 48% of packed worldlines are entirely odd-frequency, and 82.7% of all accepted frequencies are odd, both flat in T above $\sim$50. The pairwise-coprimality rule permits at most one even frequency per worldline, so the population splits into a *cold* component (all odd, exactly at rest at the turnaround) and a *mobile* one (one even axis) — and motion at maximum expansion is always single-axis.

The selection is dynamical, not just distributional: resolving acceptance by packing stage (Fig. 6), even frequencies are accepted at 20–27% across the band while the universe is sparse, but among the final quartile of survivors before jamming they are squeezed to 3–13%. What still fits into the last gaps is overwhelmingly odd — the objects that hold still at the moment the universe is largest.

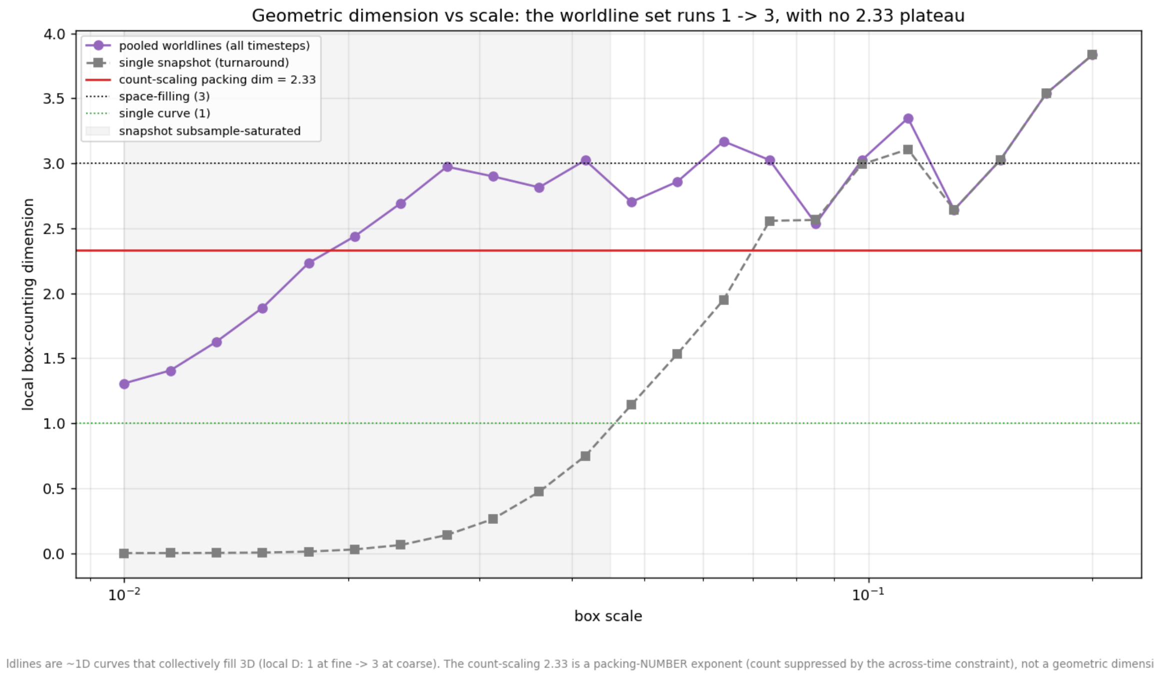

Figure 4: Local box-counting dimension vs scale for the pooled worldline bundle (3+1, $T = 200$, all timesteps pooled, three seeds). The local dimension sweeps from ≈1.3 near the packing cell (resolving individual smooth curves) through 2.76 at intermediate scales to 2.89 at coarse ones, crossing the count exponent 2.33 (line) only in passing. No scale exhibits a plateau at D.

7.2 A parameter-free speed law, and how jamming bends it

The phase freedom of even frequencies turns the mobile component's kinematics into a closed-form prediction. At the turnaround ($\sin z = 1$, $\cos z = 0$) the peculiar velocity of an axis reduces to

$$v \;=\; \big|\, a\, b \cos(b\pi/2 + f) \,\big| \;=\; |\cos f| \quad \text{for even } b \qquad (\text{and } v = 0 \text{ for odd } b), \tag{3}$$

using the rapidity constraint $|ab| = 1$. The free comoving term drops out of the velocity entirely — position and speed are independent by construction. With the proposal phases uniform, $f \sim U[0, \pi)$, the speed of a mover is arcsine-distributed:

$$P(v) \;=\; \frac{2}{\pi\sqrt{1 - v^2}}, \qquad v \in [0, 1), \qquad \langle v \rangle = \tfrac{2}{\pi}, \quad \langle v^2 \rangle = \tfrac{1}{2}. \tag{4}$$

This is a parameter-free law, and a physically suggestive one: the density diverges (integrably) at the speed cap $v = 1$, a built-in relativistic pile-up, while the cap itself is never exceeded.

The packed state tells a sharper story (Fig. 7, left). The measured mover distribution keeps the arcsine shape qualitatively — including the pile-up against the cap — but tilts: slow movers are overrepresented and the cap spike is attenuated, with

$$\langle v \rangle_{\text{packed}} = 0.539 \;\; (\text{proposal } 0.637), \qquad \langle v^2 \rangle_{\text{packed}} = 0.404 \;\; (\text{proposal } 0.500).$$

Nothing in the acceptance rule references velocity; the tilt means that *jamming itself selects on phase.* A fast mover (f near 0 or π) sweeps more comoving ground around the turnaround

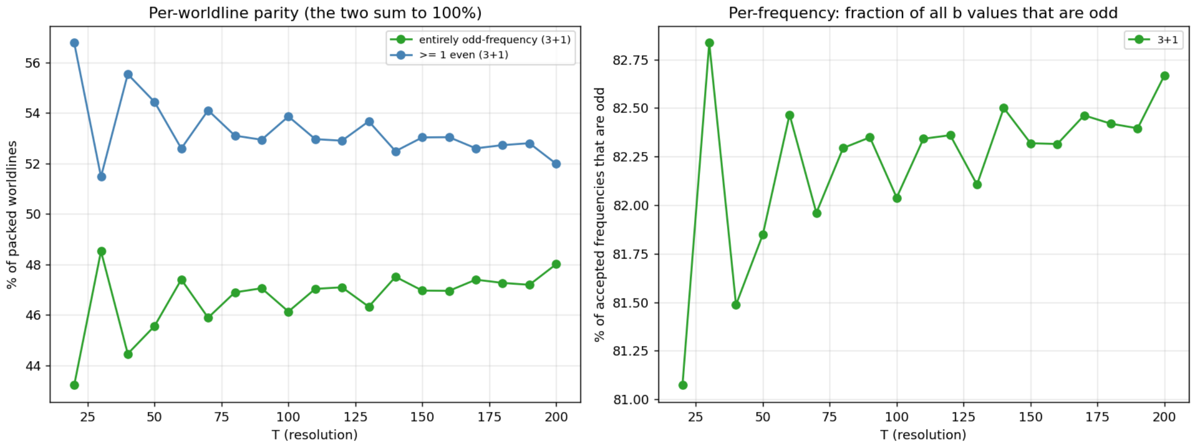

Figure 5: Composition of the jam vs resolution (3+1). Left: fraction of worldlines entirely odd-frequency vs carrying an even frequency. Right: fraction of all accepted frequencies that are odd. Both are flat above $T \approx 50$.

and is correspondingly harder to thread through a crowded universe, so the survivors' phases concentrate toward quadrature. This is selection acting on a degree of freedom invisible to any single-time snapshot — and it is resolution-independent: the suppression factor is the same at every T measured.

7.3 The equation of state

Assigning worldlines an energy E and using the kinetic pressure of the mobile component, $w = P/\rho = \frac{1}{3}\langle Ev^2\rangle/\langle E\rangle$, the turnaround equation of state is (Fig. 7, right):

$$w \ = \ 0.070, \qquad \text{flat over the full ladder } (T = 20\text{–}200),$$

computed under $E \propto \sum_k b_k$ (a frequency/"wave" dictionary) and reproduced to 1% under $E \propto$ the arc length of the spatial path in *physical* coordinates, $\int_0^\pi |d\mathbf{x}/dz|\, dz$ with $dx_i/dz = a_i b_i \cos(b_i z + f_i) + a_{1,i} \cos z$ (a "string" dictionary — note the Hubble-flow term contributes, so comoving-static worldlines also carry length): 0.0692 vs 0.0700 at $T = 200$. Both dictionaries are parity-blind — cold and mobile worldlines receive comparable energies under each — and for any parity-blind E the arcsine law predicts $w = \frac{1}{6} \times$ (mobile fraction) ≈ 0.087; the measured w sits 20% below it, which is precisely the phase-selection tilt of Section 7.2 expressed thermodynamically. The universe at maximum expansion is matter-like ($w \ll 1/3$), its pressure is carried by a single-axis-moving minority whose speeds follow a tilted arcsine law, and none of these statements depends on which of the two natural mass dictionaries one adopts, nor on the resolution.

8 Discussion

8.1 What kind of object is the jam?

Assembling the measurements: the jam is spatially featureless at every moment (exactly uniform slices, no center, no boundary, no fractality), temporally featureless in comoving terms

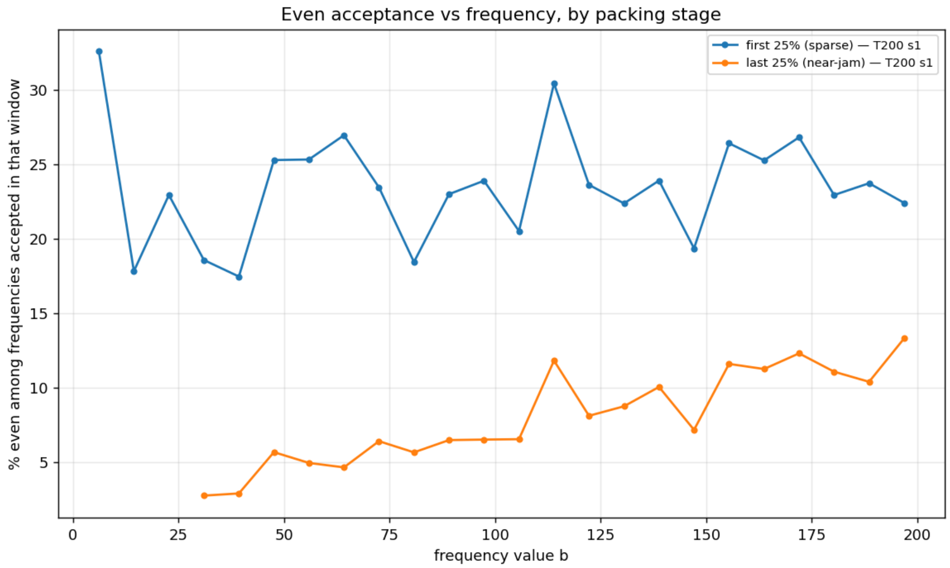

Figure 6: Even acceptance vs frequency value, resolved by packing stage (3+1, $T = 200$, a full ordered acceptance sequence of $N = 1.6 \times 10^5$ paths): first quartile of acceptances (sparse universe) vs last quartile (near jamming). The near-jam survivors skew strongly odd at every frequency.

(statistically identical slices across the loop), and yet strongly structured in every non-spatial direction — a sub-dimensional count exponent, a sharp parity composition, a phase-selected velocity law, a definite equation of state. All structure lives in the *joint, whole-loop* character of the exclusion: which trajectories can coexist, not where anything is.

Is the jam a strange attractor? The question is natural — the toolset we use (correlation integrals, generalized dimensions) comes from that field [4, 5], and the packing does exhibit attractor-like selection: iterating acceptance drives the surviving population onto a measure-zero-ish preferred subset of parameter space (odd-dominated frequencies, quadrature phases, coprime tuples) that is stable across seeds and resolutions. But the analogy has to be drawn in *parameter space*, not physical space: the spatial sets are exactly space-filling, so nothing strange lives there. What would make the reading precise is currently open: one would want a well-defined map whose iteration the RSA process approximates, evidence of sensitive dependence (do neighbouring seeds diverge in composition?), and a fractal measurement on the selected parameter-space set itself — e.g. the dimension spectrum of the accepted (b, f, a_1) population — none of which has been done. We record the hypothesis and its test surface rather than a claim.

A second reading of the count exponent deserves note: D/d is 0.78 in 3+1 and 0.69–0.70 in 2+1 at this cutoff. Whether D converges to something exact (the 3+1 value is numerically close to 7/3), and whether D/d tends to a dimension-independent constant as the model family is enlarged, are open quantitative questions an analytic treatment of a reduced case (1+1; frequency-capped bands) might settle.

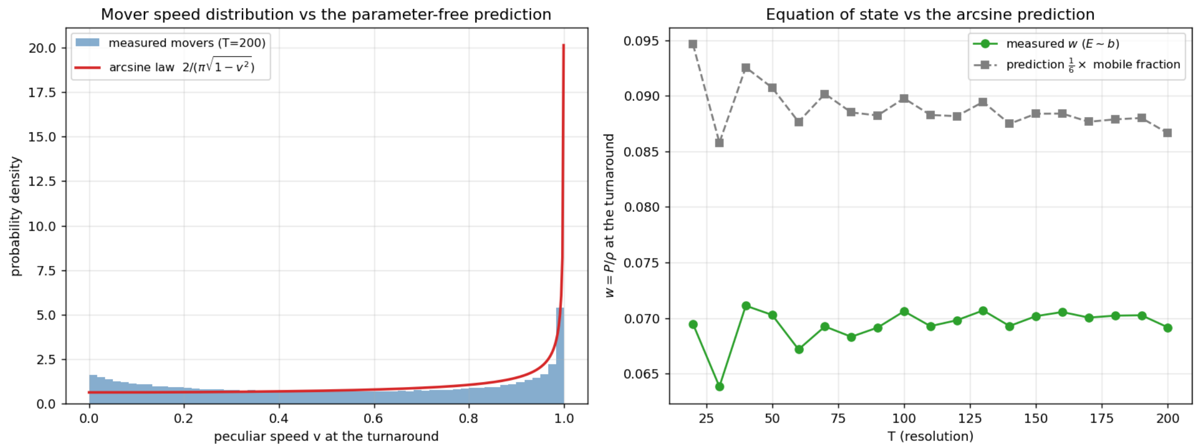

Figure 7: Left: measured turnaround speed distribution of the mobile component (3+1, $T = 200$, 1.6×10^5 movers across five seeds) against the arcsine law of the proposal measure. The relativistic pile-up at $v \to 1$ survives packing, but attenuated, with a compensating excess of slow movers: jamming selects on phase. Right: the turnaround equation of state vs resolution — flat at $w = 0.070$ across the entire ladder, ~20% below the no-selection prediction $w = \frac{1}{6} \times$ (mobile fraction).

8.2 The unexplored model

Everything above concerns the single-budget-frequency sector with singleton groups. The model as defined (Section 2) is larger in three independent directions, and we flag expectations for none of them:

1. **Multi-frequency worldlines.** The rapidity budget can be split across many components (Eq. (2)); in the limit of unrestricted splitting the family exhausts all closed unit-rapidity paths. Whether the count exponent, the parity structure, and the arcsine tilt are properties of the sector or of the budget is the sharpest next measurement.

2. **Unique groups and subpaths.** Post-jam filling by subpaths (Section 2.4) redistributes remaining capacity among existing identities — one group present in several places at once. Both the combinatorics (how much subpath capacity does a jam retain?) and the physics reading of multi-place groups are untouched.

3. **Histories.** The equation of state here is a turnaround measurement; the full $w(z)$ history of the phase model over the loop, and its behaviour approaching the bang and crunch, require a careful treatment of peculiar velocities away from $\sin z = 1$ and are left to future work.

Beyond these, the standing quantitative problems are the extrapolation from the acceptance-rate cutoff to literal jamming (a Feder-law analog for curve packings [2]), and a mechanism-level account of the two cleanest facts we cannot yet derive: the value of D, and the exactness with which jamming preserves homogeneity while selecting so strongly on everything else.

Reproducibility. Every figure is generated by committed analysis code from dumped worldline parameters, and the repository's lab notebook records each result with the commit that produced it. The packing engines, orchestration, and estimators are public at the repository above.

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